MATH 2211, SPRING 2026: MIDTERM 1 STUDY GUIDE

Just like Midterm 1, this exam will consist of 5 sections, the first of which will consist of true or false questions. For this, you will be expected to not only give the right response, but also a correct justification for that response, in one or two sentences.

Do not spend more than 8–9 minutes per problem before moving on to the next. This way you can be sure to at least *attempt* every problem in a substantial way.

While the exam will focus mainly on material covered since the first midterm, the subject is inherently cumulative, and so you should be somewhat comfortable with the stuff from earlier as well; see the study guide for Midterm 1. In addition, you should know at least the following material well:

(1) Functions and their properties:
  (a) Definition: Lecture 13
  (b) One-to-one or injective; onto or surjective: Lecture 13
  (c) Composition: Lecture 13
  (d) Bijectivity and existence of inverse function: Lecture 13.
(2) Linear transformations:
  (a) Definition and basic properties: Lecture 14
  (b) Invertibility: Lecture 14
  (c) Linear transformations with F^n as domain: Lecture 14
  (d) Linear transformations from F^n to F^m are given by $n \times m$-matrices: Lecture 15
  (e) Invertible linear transformations or isomorphisms: Lecture 16
  You should be very familiar with especially part (d)
(3) Kernel and image:
  (a) Definitions: Lecture 18.
  (b) Finding kernel and image using row reduction; reduced row echelon form: Lecture 18
  (c) Rank-nullity: Lecture 18
  (d) Injectivity and vanishing of kernel: Fact in Lecture 19
  (e) Criteria for a linear transformation between vector spaces of the same dimension to be invertible: Lecture 19
(4) Matrix multiplication:
  (a) Definition of Av: Lecture 16
  (b) Definition of AB when A is $n \times p$ and B is $p \times m$ and relationship with composition of linear transformations: Homework 5, Problem 7
  (c) See more detailed description of matrix products in terms of dot products and transposed row vectors in Lecture 20
  (d) Invertible matrices and matrix inverses; deciding invertibility and finding the inverse using RREF: Homework 5, Problem 1
(5) Bases, linear transformations and matrices:
  (a) Connection between bases for V and isomorphisms $F^n \xrightarrow{\simeq} V$: Lecture 16 and Homework 5, Problem 8
  (b) Going from linear transformations to matrices via bases: $T \mapsto [T]_{\mathcal{B}_1,\mathcal{B}_2}$. See Lecture 16. The main case we will be concerned with is when $V = W$ and $\mathcal{B} = \mathcal{B}_1 = \mathcal{B}_2$, when we just write $[T]_{\mathcal{B}}$.
  (c) Change of basis: Going from $[T]_{\mathcal{B}}$ to $[T]_{\mathcal{B}'}$ via *conjugation* by a change of basis matrix. Initial discussion is in Lecture 17, and more detailed discussion is in Lecture 23
(6) The determinant and its properties:
  (a) The 2×2 determinant and its connection with invertibility: Proposition in Lecture 21
  (b) Key properties of 2×2 determinant: Lecture 23
  (c) Uniqueness of the general $n \times n$ determinant: Proposition in Lecture 24
  (d) Behavior of determinant under row operations: Lecture 24

  (e) Computing the determinant using row operations: Lecture 24
  (f) Compatibility of determinant with products: Lecture 24 and Homework 8, Problem 2
  (g) Elementary matrices and transpose invariance of determinant: Lecture 25

(7) Diagonalizability:
  (a) Diagonal matrices and their products: Lecture 20
  (b) Definition of diagonalizable linear transformation: Lecture 20
  (c) Characteristic polynomial and eigenvalues for 2×2 matrices: Lecture 22
  (d) Deciding if a matrix is diagonalizable using its eigenvalues: Lecture 22
  (e) If matrix is diagonalizable, using appropriate basis to compute its powers: Lecture 23 and Homework 7, Problem 4.